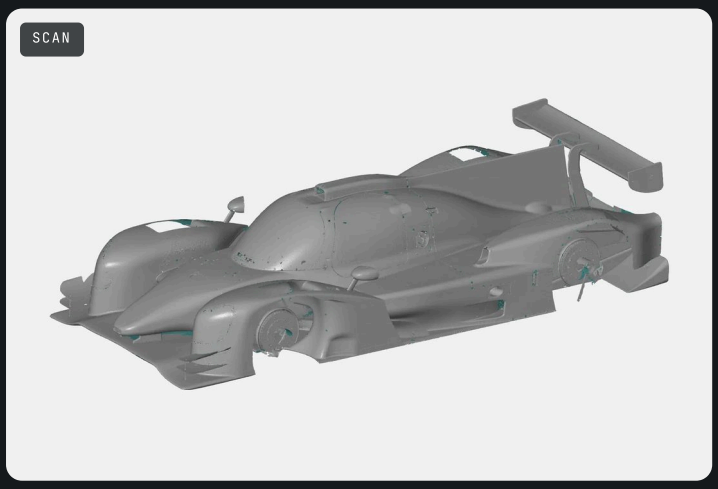
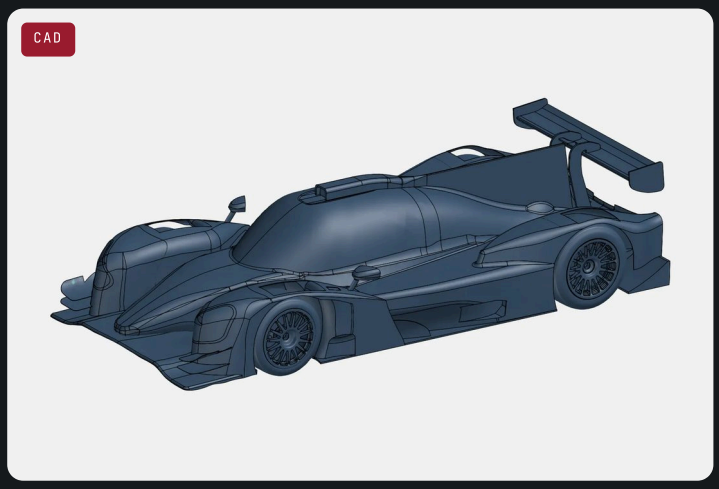
PORTFOLIO

3D Scanning & Reverse Engineering

Worn, legacy and undocumented parts captured with high-accuracy 3D scanning and rebuilt as editable, manufacturable CAD — delivered as STEP and other neutral formats.





SHOWN: PROTOTYPE RACE CAR BODY

36	8	0.02 mm	IN · AE
PROJECTS IN THIS PORTFOLIO	INDUSTRIES	SCANNER PRECISION, UP TO	ON-SITE SCANNING

01 What we deliver

Turn scan data into fully editable, manufacturable CAD.

We rebuild physical parts as parametric CAD models — for re-manufacturing legacy parts with no drawings, modifying existing designs, or creating moulds and tooling. You choose how faithful the model should be to the scanned part.

Design-intent modelling

A clean CAD model with manufacturing imperfections removed — ideal for re-manufacture from 3D models or 2D drawings.

NURBS surface models

Precise freeform surfaces fitted over scan data where an exact as-is representation matters.

Hybrid modelling

Prismatic features modelled to design intent, freeform areas surfaced to the scan — the best of both.

Drawings on request

2D manufacturing drawings with GD&T generated from the final model.

02 Deliverables

- Parametric CAD (.STEP, .IGES, .X_T)
- Scan data (.STL)
- 2D drawings (.PDF, .DWG)
- Deviation report vs. scan

03 How we work

Scan to solution in four steps.

STEP 01

Capture

We scan the part, product or site — in our lab or on location.

STEP 02

Process

Scans are aligned, cleaned and merged into usable, watertight data.

STEP 03

Engineer

CAD, Class A surfaces, BIM or inspection built from that data.

STEP 04

Deliver

CAD files, drawings and reports in the formats your team uses.

04 Industries in this portfolio

- Automobile 17
- Consumer Products 7
- Industrial 4
- Aerospace 2
- Mining 2
- Marine 2
- Oil & Gas 1
- Healthcare 1

05 Common questions

What do I need to send to get a quote?

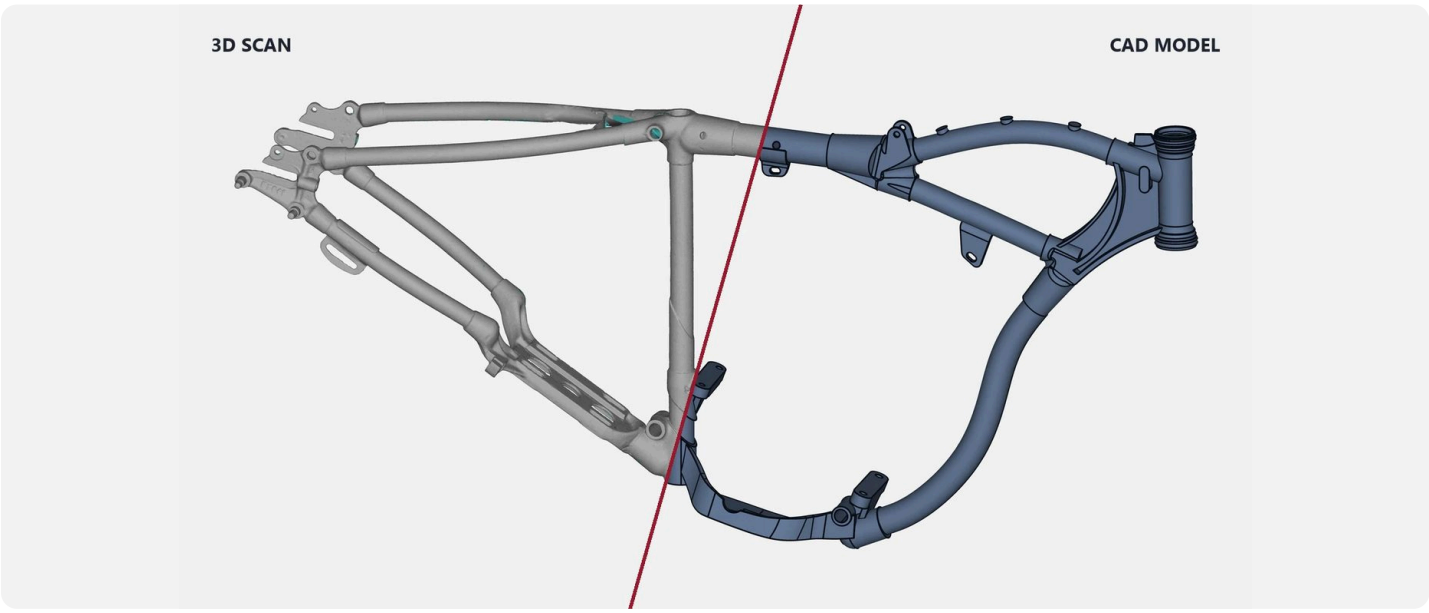
Clear photos of the part from several angles, approximate dimensions, the quantity you plan to make, and what you need delivered (CAD only, CAD and drawings, or inspection). If you have an old drawing, even a partial one, send that too.

Can you reverse engineer a broken or worn part?

Yes. We scan what remains and rebuild the missing or worn geometry using design intent, symmetry, mating parts and any information you can share.

Will the new part be identical to the original?

It will match the original design intent and fit. Where wear has changed the shape, we restore the original geometry rather than copying the wear.



PROJECTS **01–04 of 36**

Real parts scanned and engineered by the VShark team.

01 Prototype Race Car Body

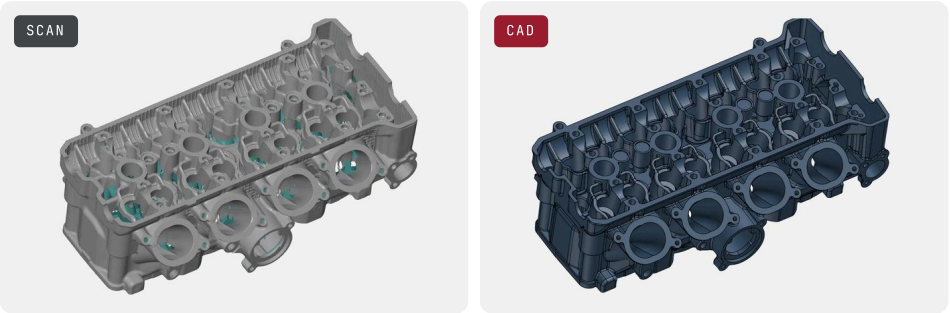
AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING · DIGITAL METROLOGY



Full body of a prototype race car scanned, rebuilt as CAD and checked against the scan.

02 Engine Cylinder Head

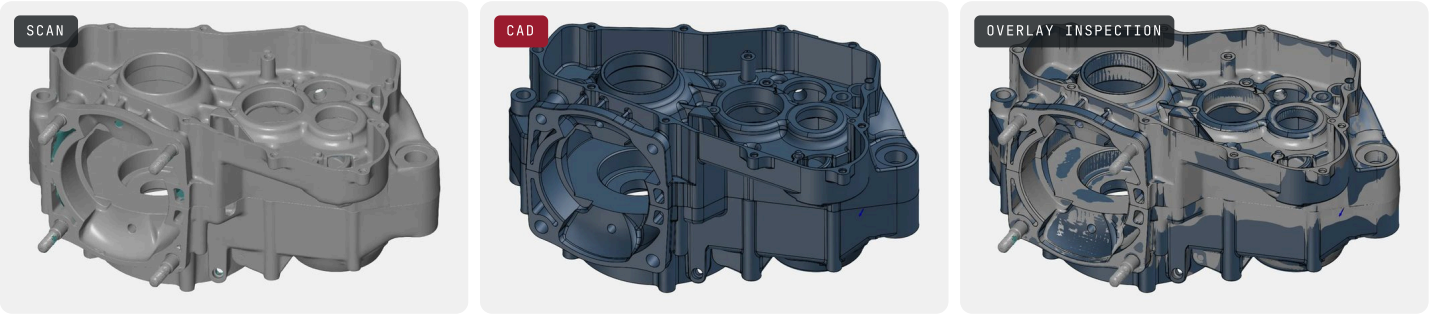
AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING



Engine cylinder head scanned and rebuilt as parametric CAD.

03 Engine Crankcase Casting

AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING · DIGITAL METROLOGY



Complex crankcase casting digitised and remodelled, with a deviation overlay.

04 Helicopter Fuselage

AEROSPACE · 3D SCANNING · REVERSE ENGINEERING

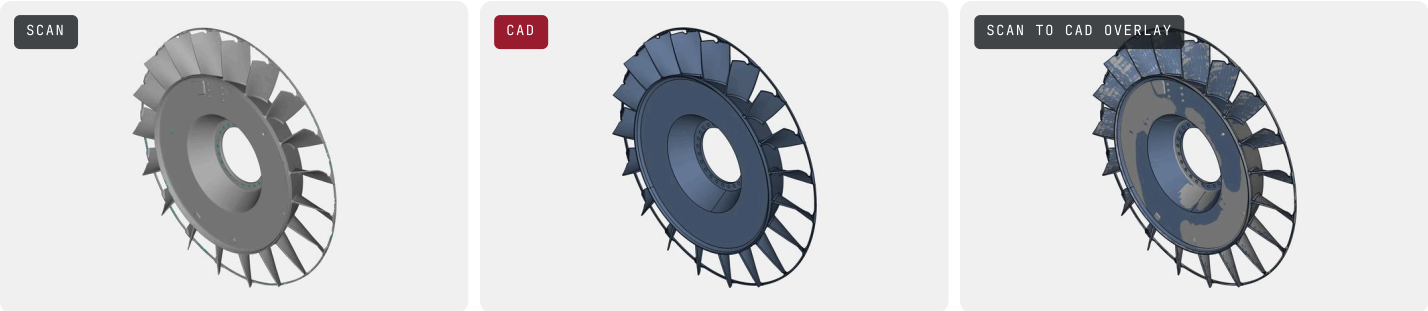


Helicopter fuselage scanned and rebuilt as editable CAD surfaces.

Real parts scanned and engineered by the VShark team.

05 Large Industrial Fan Impeller

INDUSTRIAL · 3D SCANNING · REVERSE ENGINEERING



Large fan impeller digitised and remodelled as editable CAD for re-manufacture.

06 Steering Wheel Armature

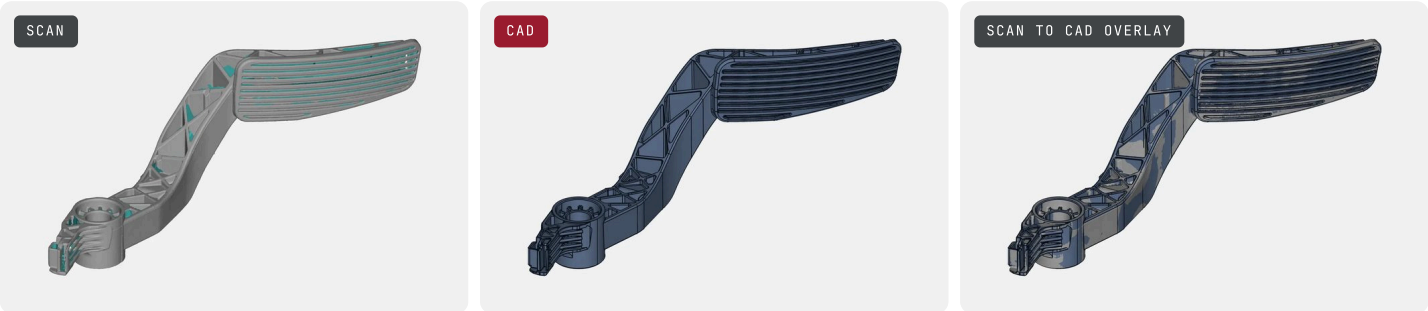
AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING



Steering wheel armature captured and rebuilt as manufacturable CAD.

07 Accelerator Pedal

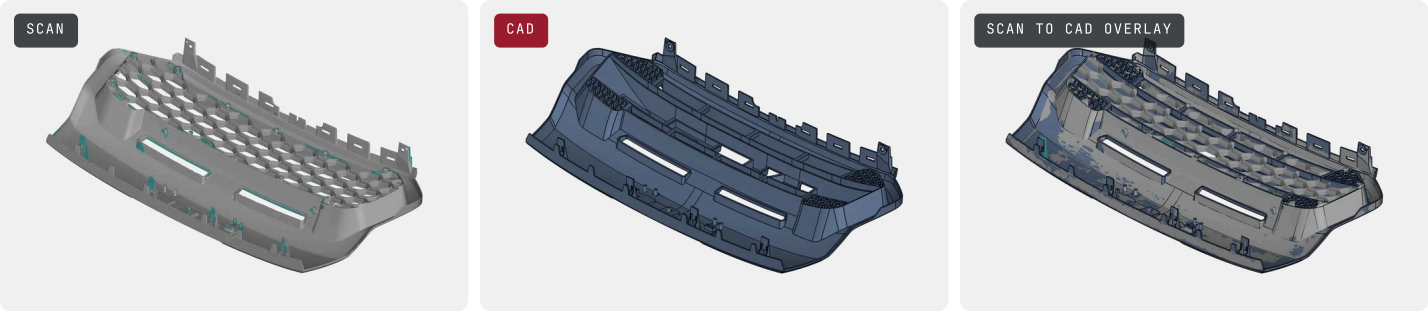
AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING



Accelerator pedal scanned and modelled as editable CAD.

08 Front Grille

AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING



Front grille digitised and rebuilt as editable CAD.

Real parts scanned and engineered by the VShark team.

09 Grille Insert

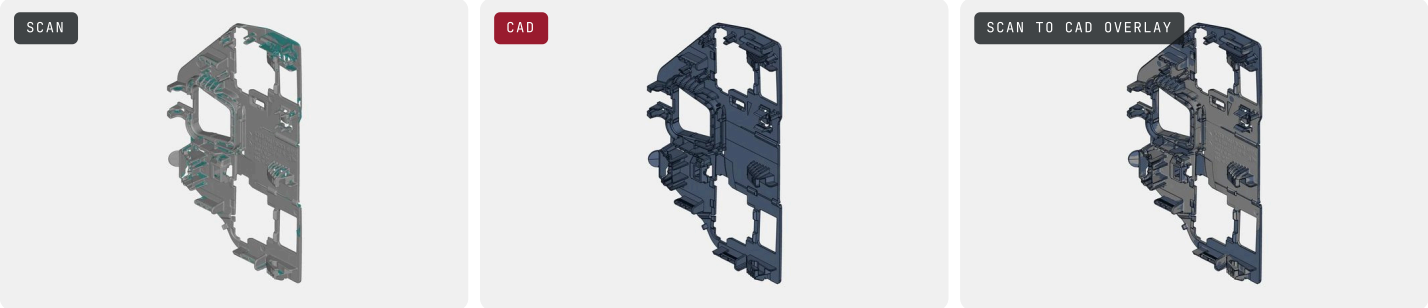
AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING



Grille insert captured and remodelled to design intent.

10 Plastic Carrier Panel

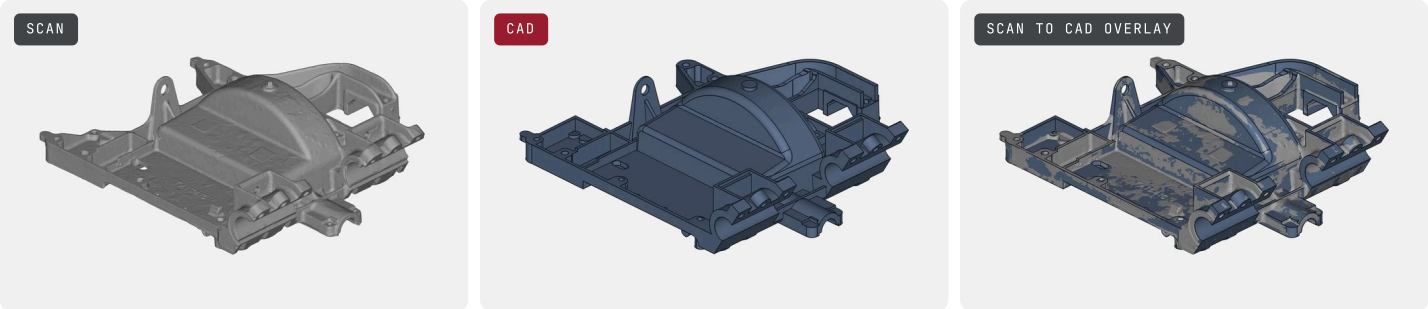
AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING



Moulded plastic carrier panel scanned and rebuilt as production CAD.

11 Transmission Lower Housing Casting

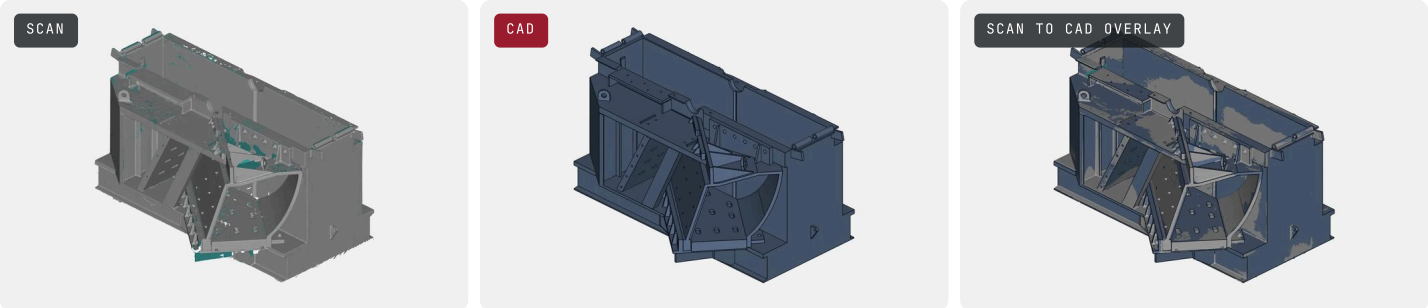
AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING · DIGITAL METROLOGY



Transmission housing casting scanned, modelled and compared against the CAD.

12 Crusher Main Throat

MINING · 3D SCANNING · REVERSE ENGINEERING

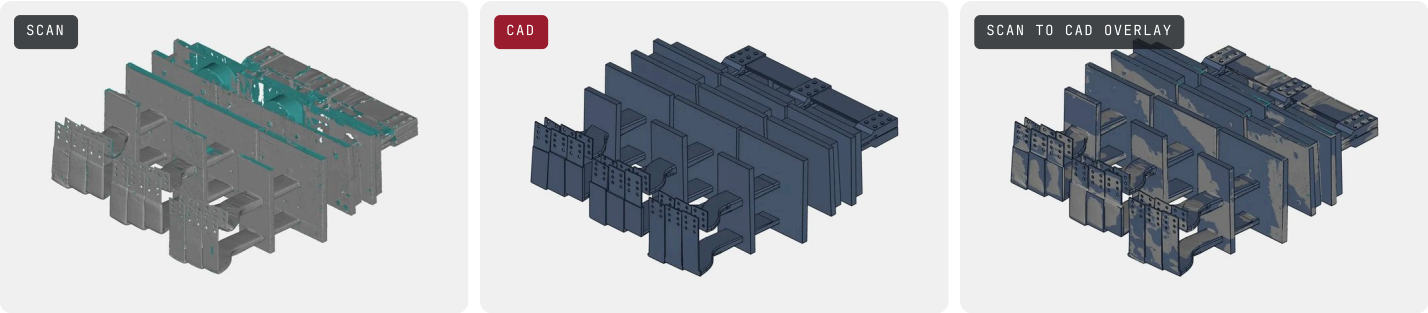


Crusher main throat scanned and rebuilt as CAD so replacements can be produced.

Real parts scanned and engineered by the VShark team.

13 Shorting Bus Frame

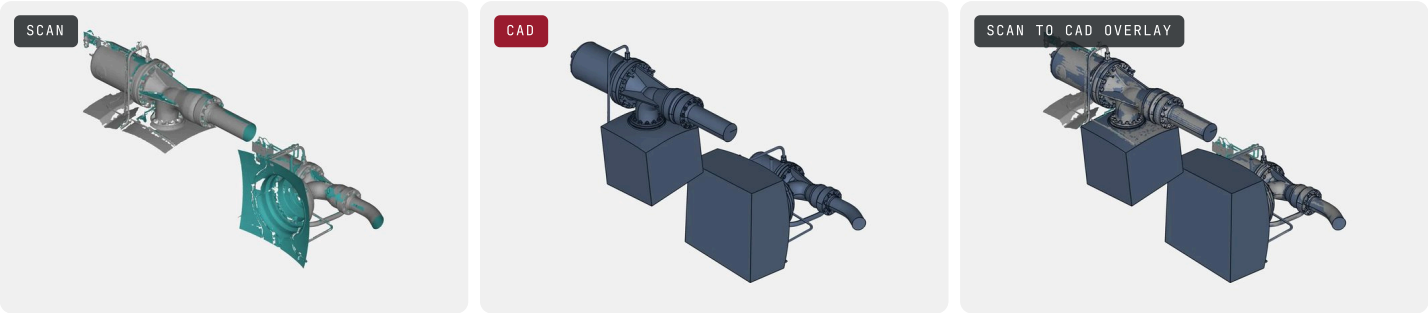
INDUSTRIAL · 3D SCANNING · REVERSE ENGINEERING



Electrical shorting bus frame captured and modelled as editable CAD.

14 Pump and Valve Assembly

OIL & GAS · 3D SCANNING · REVERSE ENGINEERING



Pump and valve assembly digitised and modelled as editable CAD.

15 Extruder Screw

INDUSTRIAL · 3D SCANNING · REVERSE ENGINEERING



Extruder screw scanned and remodelled as editable CAD.

16 Safety Helmet Shell

CONSUMER PRODUCTS · 3D SCANNING · REVERSE ENGINEERING



Safety helmet shell scanned and rebuilt as smooth production CAD.

Real parts scanned and engineered by the VShark team.

17 Plastic Jerry Can

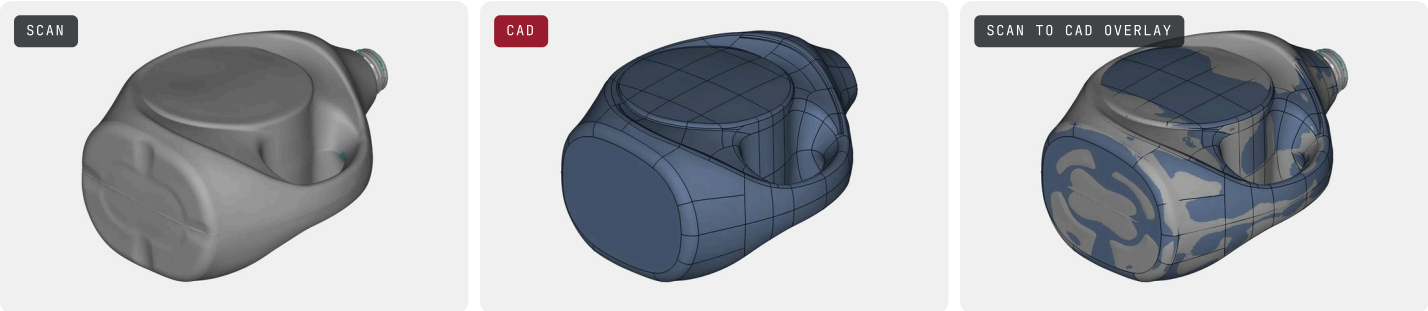
CONSUMER PRODUCTS · 3D SCANNING · REVERSE ENGINEERING



Plastic jerry can captured and rebuilt as production CAD.

18 Blow-Moulded Plastic Can

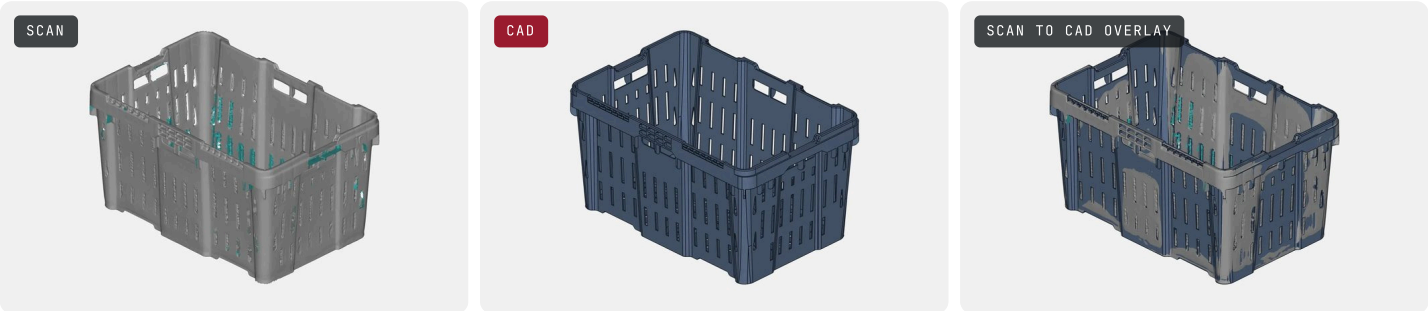
CONSUMER PRODUCTS · 3D SCANNING · REVERSE ENGINEERING



Blow-moulded can with handle scanned and rebuilt as editable CAD.

19 Vented Plastic Crate

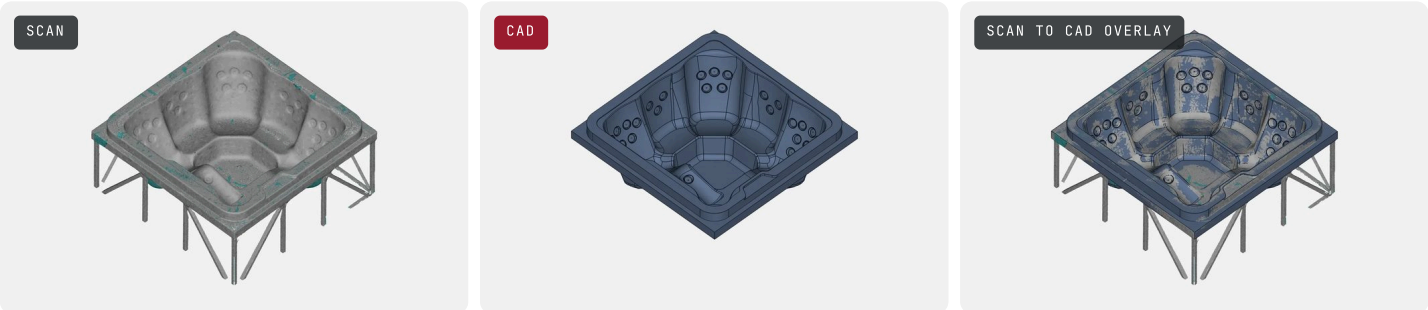
CONSUMER PRODUCTS · 3D SCANNING · REVERSE ENGINEERING



Vented plastic crate digitised and rebuilt as production CAD.

20 Spa Hot Tub Shell

CONSUMER PRODUCTS · 3D SCANNING · REVERSE ENGINEERING

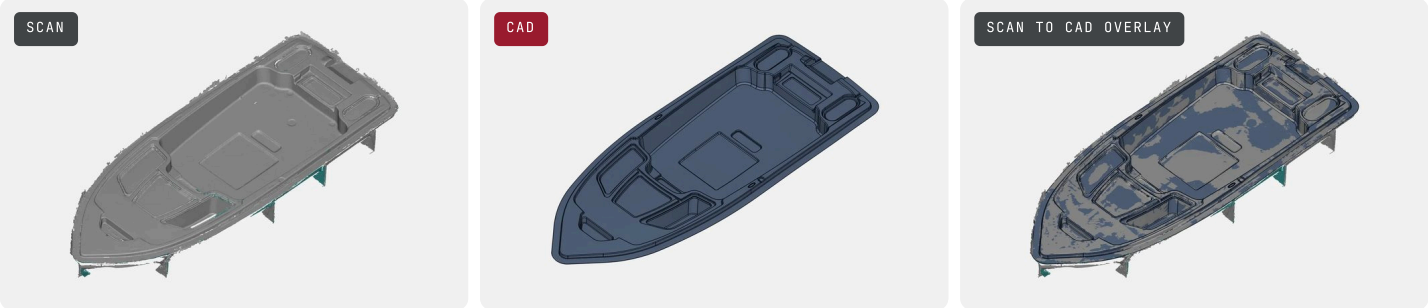


Spa hot tub shell scanned and rebuilt as smooth CAD surfaces.

Real parts scanned and engineered by the VShark team.

21 Boat Deck

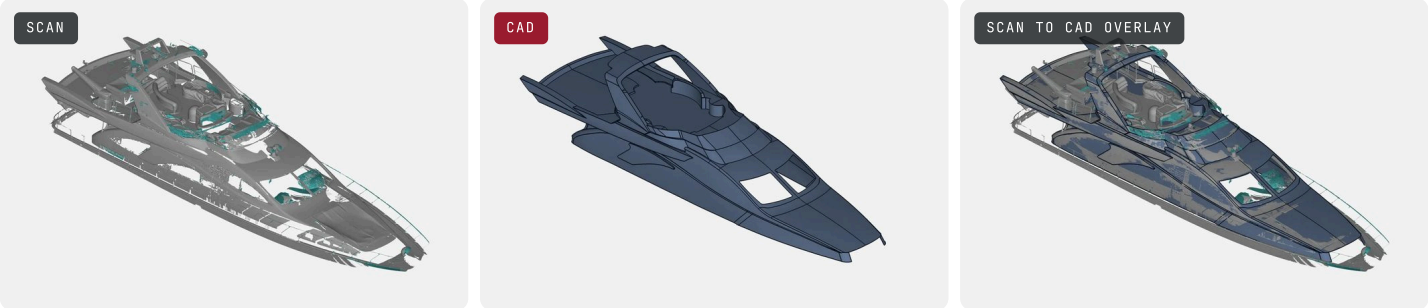
MARINE · 3D SCANNING · REVERSE ENGINEERING



Boat deck scanned and modelled as editable CAD.

22 Motor Yacht

MARINE · 3D SCANNING · REVERSE ENGINEERING



Motor yacht hull and superstructure scanned and modelled as CAD.

23 Electric Dirt Bike Frame

AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING · DIGITAL METROLOGY



Whole bike scanned; the frame extracted and rebuilt as editable CAD.

24 Car Body-in-White

AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING

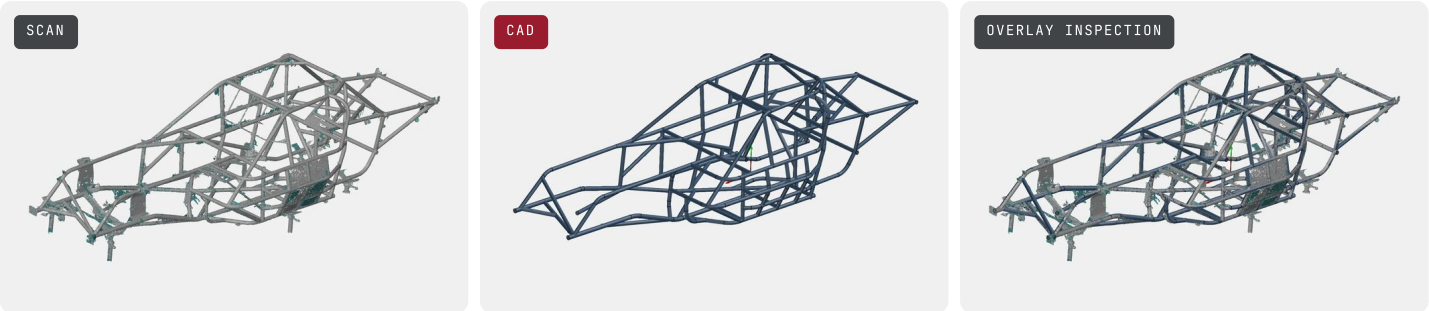


Complete body-in-white shell digitised and converted to CAD.

Real parts scanned and engineered by the VShark team.

25 Tubular Roll Cage

AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING · DIGITAL METROLOGY



Space-frame roll cage captured tube by tube and modelled for fabrication.

26 Truck Chassis Frame

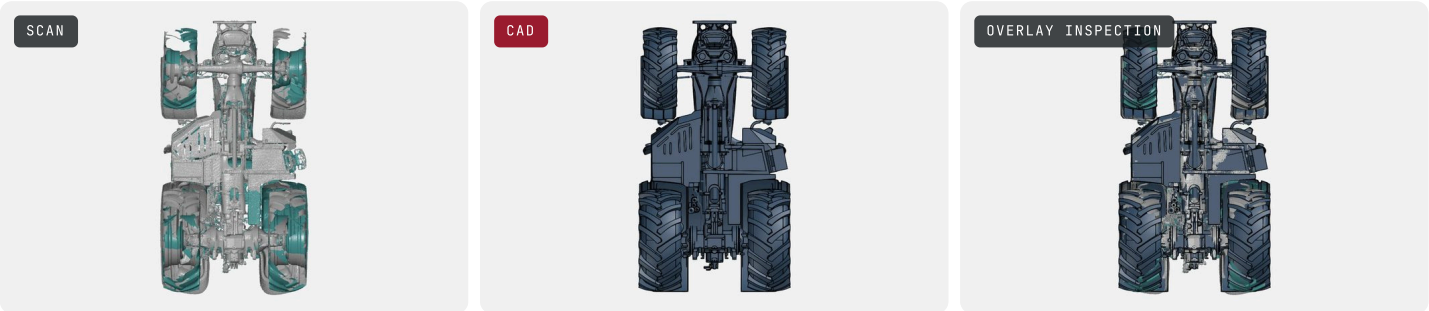
AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING · DIGITAL METROLOGY



Heavy truck chassis scanned and rebuilt as a CAD assembly.

27 Tractor Underbody

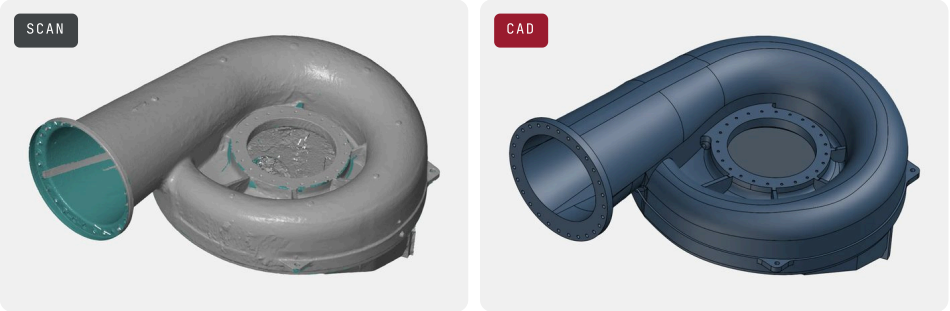
AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING · DIGITAL METROLOGY



Tractor underbody digitised for redesign and packaging studies.

28 Centrifugal Pump Volute

INDUSTRIAL · 3D SCANNING · REVERSE ENGINEERING

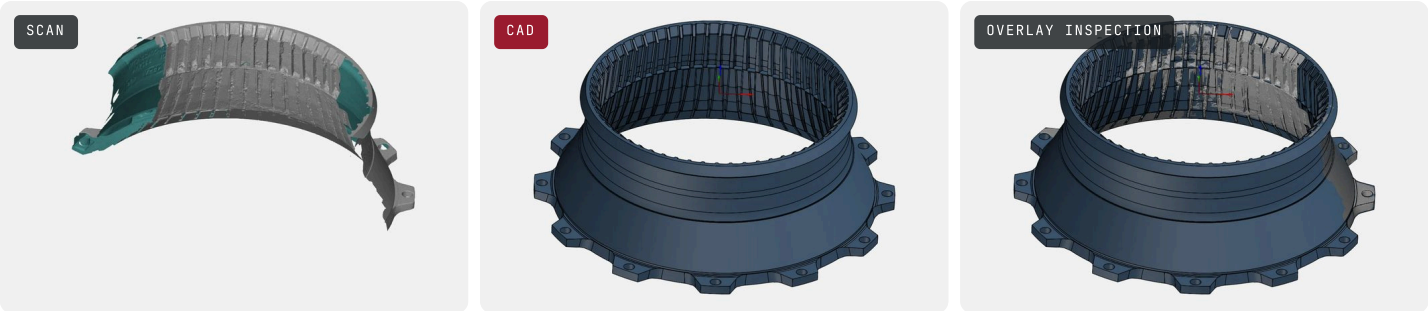


Pump volute casing reverse-engineered for re-manufacture.

Real parts scanned and engineered by the VShark team.

29 Cone Crusher Liner

MINING · 3D SCANNING · REVERSE ENGINEERING · DIGITAL METROLOGY



Wear liner scanned and modelled so replacements can be produced.

30 Smartphone Back Housing

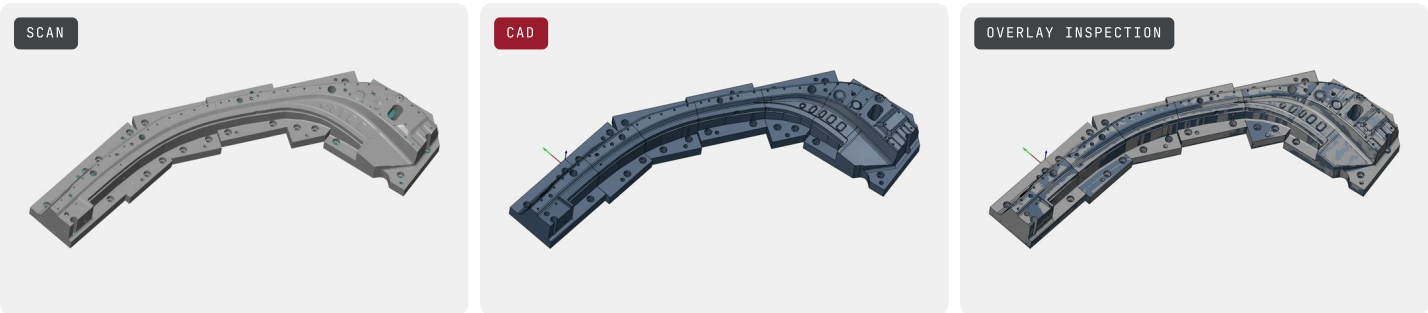
CONSUMER PRODUCTS · 3D SCANNING · REVERSE ENGINEERING · DIGITAL METROLOGY



Thin-walled consumer housing captured and remodelled to design intent.

31 Automotive Mold Insert

AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING · DIGITAL METROLOGY



Mold insert assembly reverse-engineered for tool repair and duplication.

32 Plastic Shopping Trolley

CONSUMER PRODUCTS · 3D SCANNING · REVERSE ENGINEERING · DIGITAL METROLOGY

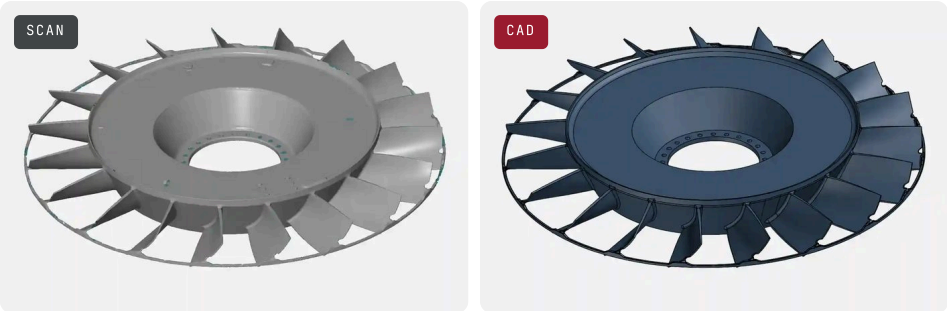


Large moulded trolley scanned and rebuilt as production CAD.

Real parts scanned and engineered by the VShark team.

33 Impeller

AEROSPACE · 3D SCANNING · REVERSE ENGINEERING



Bladed impeller captured and modelled with precise freeform blade surfaces.

34 Bumper

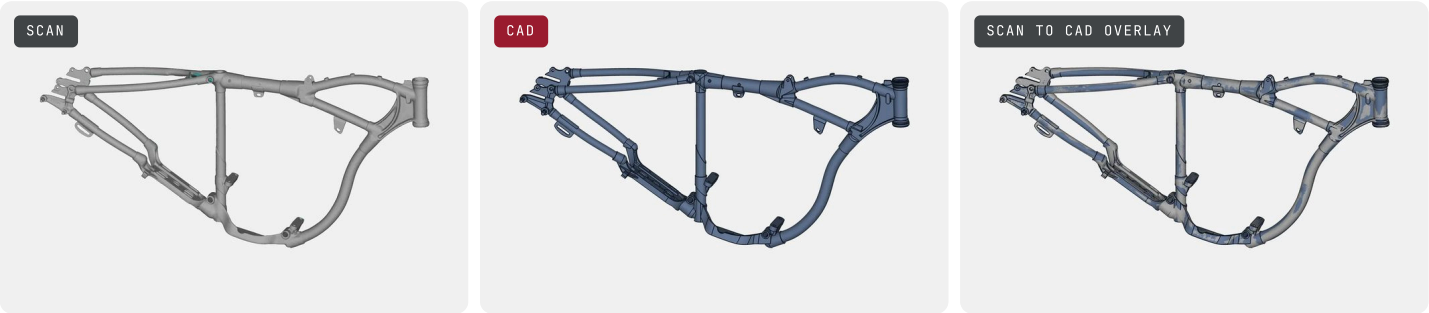
AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING · CLASS A SURFACING



Vehicle bumper scanned and resurfaced for tooling.

35 Motorcycle Tubular Frame

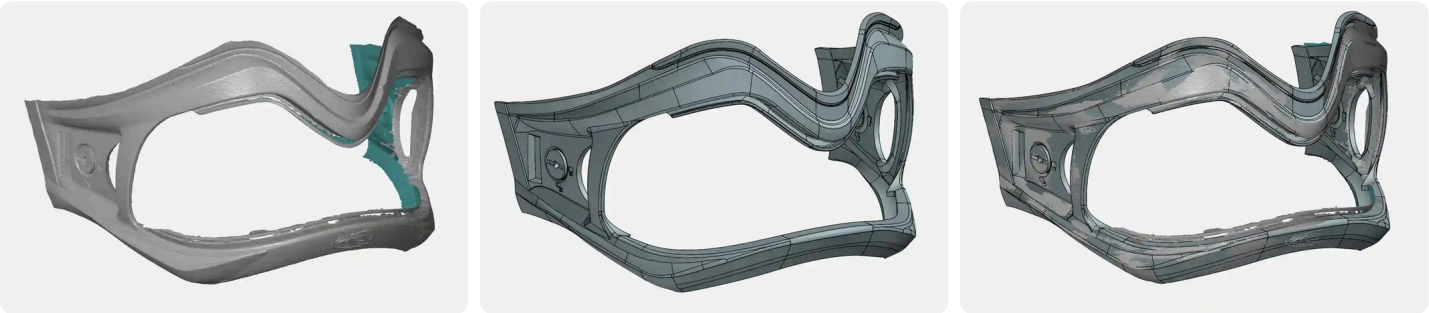
AUTOMOBILE · 3D SCANNING · REVERSE ENGINEERING



Tubular motorcycle frame digitised and converted into an editable CAD model.

36 Eyewear Frame

HEALTHCARE · 3D SCANNING · REVERSE ENGINEERING



Eyewear frame surfaced over scan data for a custom-fit product.

VSHARK SOLUTIONS PVT LTD

Have a part with no drawing?

Send clear photos from several angles, approximate dimensions, the quantity you plan to make and what you need delivered: CAD only, CAD and drawings, or inspection. If you have an old drawing, even a partial one, send that too.

We usually reply within one working day. NDAs on request — your files are used only for your project.

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